

Student:Email:
Prakriti Dhungelprakriti.dhungel98@gmail.com

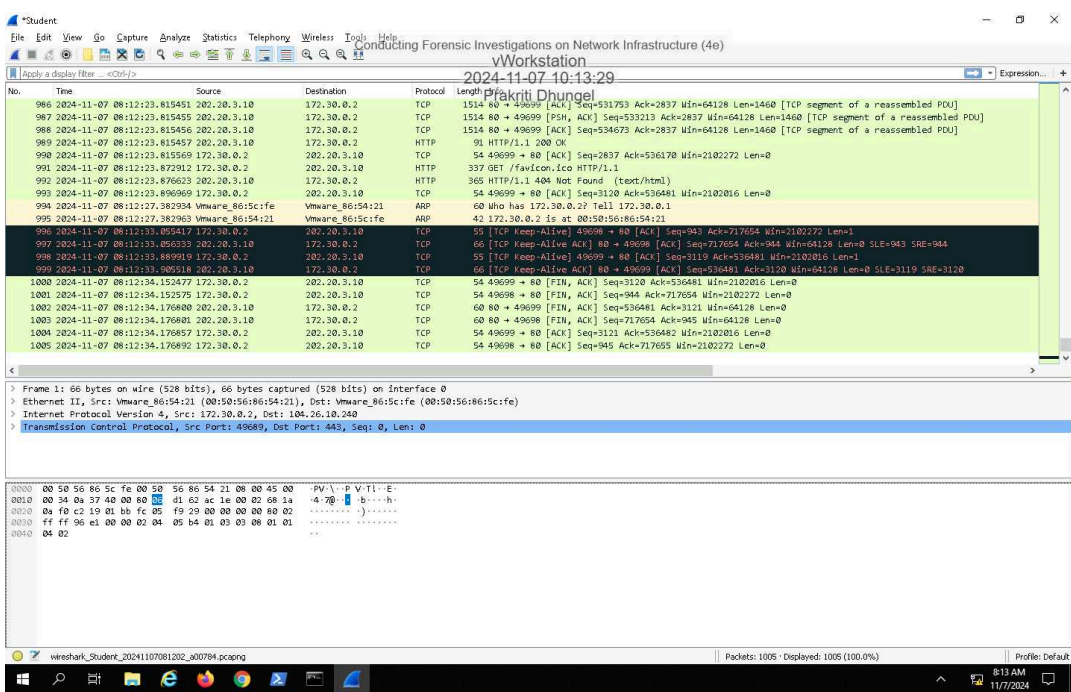
Time on Task:Progress:
2 hours, 24 minutes100%

Report Generated: Thursday, November 7, 2024 at 12:23 PM

Section 1: Hands-On Demonstration

Part 1: Perform Packet Capture and Analysis

11. Make a screen capture showing the timestamp-sorted traffic.



Conducting Forensic Investigations on Network Infrastructure (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 09

13. Make a screen capture showing the IP-filtered traffic.

Wireshark screenshot showing IP-filtered traffic. The filter bar at the top is set to `ip.addr == 202.20.3.10`. The packet list shows 26 packets, all of which are filtered. The packet details pane shows the selected packet (No. 13) with its Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol headers. The packet bytes pane shows the raw data in hexadecimal and ASCII.

15. Make a screen capture showing the port-filtered traffic.

Wireshark screenshot showing port-filtered traffic. The filter bar at the top is set to `tcp.port == 80`. The packet list shows 35 packets, all of which are filtered. The packet details pane shows the selected packet (No. 13) with its Ethernet II, Internet Protocol Version 4, and Transmission Control Protocol headers. The packet bytes pane shows the raw data in hexadecimal and ASCII.

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17. Make a screen capture showing the TCP push flag-filtered traffic.

The screenshot shows Wireshark with the filter `tcp.flags.push == 1` applied. The packet list displays several TCP segments with the PUSH flag set. The packet details pane shows the Hypertext Transfer Protocol section.

No.	Time	Source	Destination	Protocol	Length	Info
12	2024-11-07 08:12:22.369658	172.30.0.2	202.28.3.10	HTTP	386	GET / / HTTP/1.1
16	2024-11-07 08:12:22.374740	202.28.3.10	172.30.0.2	HTTP	355	HTTP/1.1 200 OK (text/html)
18	2024-11-07 08:12:22.685360	172.30.0.2	202.28.3.10	HTTP	371	GET /Font-Awesome-4.7/css/font-awesome.min.css HTTP/1.1
20	2024-11-07 08:12:22.688785	172.30.0.2	202.28.3.10	HTTP	351	GET /css/bootstrap.min.css HTTP/1.1
25	2024-11-07 08:12:22.610392	202.28.3.10	172.30.0.2	TCP	1514	80 → 49699 [PSH, ACK] Seq=5841 Ack=318 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
30	2024-11-07 08:12:22.610396	202.28.3.10	172.30.0.2	TCP	1514	80 → 49699 [PSH, ACK] Seq=13141 Ack=318 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
43	2024-11-07 08:12:22.611785	202.28.3.10	172.30.0.2	TCP	1514	80 → 49699 [PSH, ACK] Seq=27741 Ack=318 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
45	2024-11-07 08:12:22.611787	202.28.3.10	172.30.0.2	TCP	634	HTTP/1.1 200 OK (text/css)
50	2024-11-07 08:12:22.611711	202.28.3.10	172.30.0.2	TCP	1514	80 → 49698 [PSH, ACK] Seq=2062 Ack=630 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
57	2024-11-07 08:12:22.616984	202.28.3.10	172.30.0.2	TCP	1514	80 → 49698 [PSH, ACK] Seq=16362 Ack=630 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
66	2024-11-07 08:12:22.630662	202.28.3.10	172.30.0.2	TCP	1514	80 → 49698 [PSH, ACK] Seq=28042 Ack=630 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
72	2024-11-07 08:12:22.630669	202.28.3.10	172.30.0.2	TCP	1514	80 → 49698 [PSH, ACK] Seq=36802 Ack=630 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
86	2024-11-07 08:12:22.631231	202.28.3.10	172.30.0.2	TCP	1514	80 → 49698 [PSH, ACK] Seq=55782 Ack=630 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
88	2024-11-07 08:12:22.634089	172.30.0.2	202.28.3.10	HTTP	354	GET /css/template-style.css HTTP/1.1
103	2024-11-07 08:12:22.636353	202.28.3.10	172.30.0.2	TCP	1514	80 → 49698 [PSH, ACK] Seq=76222 Ack=630 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
106	2024-11-07 08:12:22.636356	202.28.3.10	172.30.0.2	TCP	1514	80 → 49698 [PSH, ACK] Seq=80602 Ack=630 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
113	2024-11-07 08:12:22.636773	202.28.3.10	172.30.0.2	TCP	1514	80 → 49698 [PSH, ACK] Seq=89362 Ack=630 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
119	2024-11-07 08:12:22.636778	202.28.3.10	172.30.0.2	TCP	1514	80 → 49698 [PSH, ACK] Seq=98122 Ack=630 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
126	2024-11-07 08:12:22.637129	202.28.3.10	172.30.0.2	TCP	1514	80 → 49698 [PSH, ACK] Seq=106882 Ack=630 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
128	2024-11-07 08:12:22.637131	202.28.3.10	172.30.0.2	TCP	1514	80 → 49698 [PSH, ACK] Seq=109802 Ack=630 Win=64128 Len=1460 [TCP segment of a reassembled PDU]
136	2024-11-07 08:12:22.638351	202.28.3.10	172.30.0.2	TCP	1514	80 → 49699 [PSH, ACK] Seq=37081 Ack=618 Win=64128 Len=1460 [TCP segment of a reassembled PDU]

Frame 12: 386 bytes on wire (3088 bits), 386 bytes captured (3088 bits) on interface 0
Ethernet II, Src: Vmware_86:54:12 (00:50:56:86:54:12), Dst: Vmware_86:5c:fe (00:50:56:86:5c:fe)
Internet Protocol Version 4, Src: 172.30.0.2, Dst: 202.28.3.10
Transmission Control Protocol, Src Port: 49698, Dst Port: 80, Seq: 1, Ack: 1, Len: 332
Hypertext Transfer Protocol

0000 00 50 56 86 5c fe 00 50 56 86 54 12 00 00 45 00 ...PV...P V T I E
0010 01 74 78 59 40 00 00 06 07 ec ac 1e 00 02 ca 14 ...XW...
0020 03 0a c2 22 00 50 99 ec cc 9a 6e 5e 3e 85 50 18 ...P...P...
0030 20 14 9b 34 00 00 47 45 54 20 2f 20 48 54 54 50 ...-4-GE T / HTTP
0040 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 32 30 32 2e .../1.1-Host: 202.
0050 32 30 3e 33 2e 31 30 0d 0a 55 73 65 72 2d 41 67 ...20.3.10 User-Ag
0060 65 6e 7a 3a 20 4d 6f 7a 69 6e 6c 61 2f 35 2e 30 ...ent: Nox flla/5.0
0070 20 28 57 69 6e 64 6f 77 73 20 4e 54 20 31 30 2e ... (window s NT 10.
0080 30 3b 20 57 69 6e 36 34 3b 20 78 36 34 3b 20 72 ...0; Win64 ; x64; r
0090 76 3a 39 31 2e 30 29 20 47 65 63 6b 6f 2f 32 30 ...v:91.0) Gecko/20
00a0 31 30 30 31 30 31 20 46 69 72 65 66 6f 78 2f 39 ...100101 Firefox/9
00b0 31 2e 30 0d 0a 61 63 63 65 70 7a 3a 20 74 65 78 ...1.0-Acc ept: tex
00c0 6f 6e 2f 78 68 74 6d 6c 2c 61 70 70 69 63 61 70 ...t/html; applicati
00d0 6f 6e 2f 78 68 74 6d 6c 2b 78 6d 6c 2c 61 70 70 ...on/xhtml+xml,app

19. Make a screen capture showing the http-filtered traffic.

The screenshot shows Wireshark with the filter `http` applied. The packet list displays several HTTP GET requests. The packet details pane shows the Hypertext Transfer Protocol section.

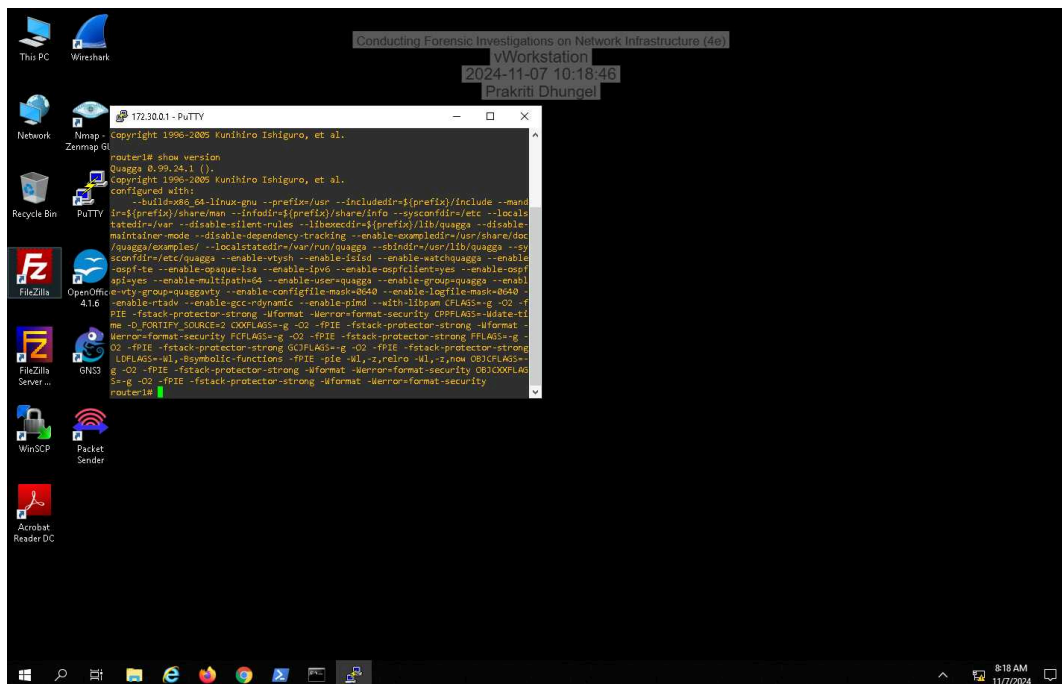
No.	Time	Source	Destination	Protocol	Length	Info
12	2024-11-07 08:12:22.369658	172.30.0.2	202.28.3.10	HTTP	386	GET / / HTTP/1.1
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18	2024-11-07 08:12:22.685360	172.30.0.2	202.28.3.10	HTTP	371	GET /Font-Awesome-4.7/css/font-awesome.min.css HTTP/1.1
20	2024-11-07 08:12:22.688785	172.30.0.2	202.28.3.10	HTTP	351	GET /css/bootstrap.min.css HTTP/1.1
45	2024-11-07 08:12:22.611787	202.28.3.10	172.30.0.2	HTTP	634	HTTP/1.1 200 OK (text/css)
88	2024-11-07 08:12:22.634089	172.30.0.2	202.28.3.10	HTTP	354	GET /css/template-style.css HTTP/1.1
137	2024-11-07 08:12:22.638352	202.28.3.10	172.30.0.2	HTTP	856	HTTP/1.1 200 OK (text/css)
149	2024-11-07 08:12:22.640541	202.28.3.10	172.30.0.2	HTTP	376	HTTP/1.1 200 OK (text/css)
151	2024-11-07 08:12:22.670638	172.30.0.2	202.28.3.10	HTTP	363	GET /images/ISSA_CorporationTechs_450a.png HTTP/1.1
179	2024-11-07 08:12:22.677387	202.28.3.10	172.30.0.2	HTTP	1300	HTTP/1.1 200 OK (PNG)
181	2024-11-07 08:12:22.680081	172.30.0.2	202.28.3.10	HTTP	351	GET /images/technical_icon.png HTTP/1.1
202	2024-11-07 08:12:22.710397	202.28.3.10	172.30.0.2	HTTP	311	HTTP/1.1 200 OK (PNG)
204	2024-11-07 08:12:22.762839	172.30.0.2	202.28.3.10	HTTP	364	GET /images/engineering-technical-18496.jpg HTTP/1.1
205	2024-11-07 08:12:22.769551	172.30.0.2	202.28.3.10	HTTP	368	GET /images/bringing_in_those_ideas_500_clr.gif HTTP/1.1
213	2024-11-07 08:12:22.771642	202.28.3.10	172.30.0.2	HTTP	1452	HTTP/1.1 200 OK (JPEG JFIF image)
662	2024-11-07 08:12:23.039971	202.28.3.10	172.30.0.2	HTTP	224	HTTP/1.1 200 OK (GIF89a)
664	2024-11-07 08:12:23.061968	172.30.0.2	202.28.3.10	HTTP	345	GET /images/tm-jm-4.jpg HTTP/1.1
698	2024-11-07 08:12:23.134923	202.28.3.10	172.30.0.2	HTTP	721	HTTP/1.1 200 OK (JPEG JFIF image)
700	2024-11-07 08:12:23.177166	172.30.0.2	202.28.3.10	HTTP	329	GET /js/shortcut.js HTTP/1.1
705	2024-11-07 08:12:23.179773	202.28.3.10	172.30.0.2	HTTP	152	HTTP/1.1 200 OK (application/javascript)
707	2024-11-07 08:12:23.217237	172.30.0.2	202.28.3.10	HTTP	375	GET /images/tm-catalyst-bg.jpg HTTP/1.1

Frame 12: 386 bytes on wire (3088 bits), 386 bytes captured (3088 bits) on interface 0
Ethernet II, Src: Vmware_86:54:12 (00:50:56:86:54:12), Dst: Vmware_86:5c:fe (00:50:56:86:5c:fe)
Internet Protocol Version 4, Src: 172.30.0.2, Dst: 202.28.3.10
Transmission Control Protocol, Src Port: 49698, Dst Port: 80, Seq: 1, Ack: 1, Len: 332
Hypertext Transfer Protocol

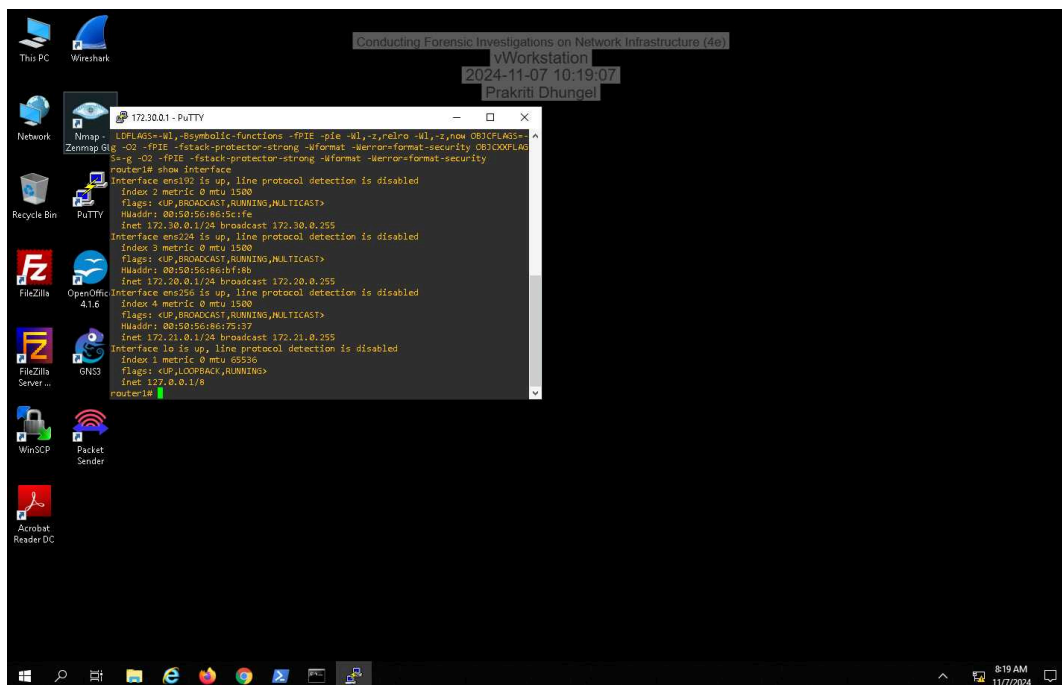
0000 00 50 56 86 5c fe 00 50 56 86 54 12 00 00 45 00 ...PV...P V T I E
0010 01 74 78 59 40 00 00 06 07 ec ac 1e 00 02 ca 14 ...XW...
0020 03 0a c2 22 00 50 99 ec cc 9a 6e 5e 3e 85 50 18 ...P...P...
0030 20 14 9b 34 00 00 47 45 54 20 2f 20 48 54 54 50 ...-4-GE T / HTTP
0040 2f 31 2e 31 0d 0a 48 6f 73 74 3a 20 32 30 32 2e .../1.1-Host: 202.
0050 32 30 3e 33 2e 31 30 0d 0a 55 73 65 72 2d 41 67 ...20.3.10 User-Ag
0060 65 6e 7a 3a 20 4d 6f 7a 69 6e 6c 61 2f 35 2e 30 ...ent: Nox flla/5.0
0070 20 28 57 69 6e 64 6f 77 73 20 4e 54 20 31 30 2e ... (window s NT 10.
0080 30 3b 20 57 69 6e 36 34 3b 20 78 36 34 3b 20 72 ...0; Win64 ; x64; r
0090 76 3a 39 31 2e 30 29 20 47 65 63 6b 6f 2f 32 30 ...v:91.0) Gecko/20
00a0 31 30 30 31 30 31 20 46 69 72 65 66 6f 78 2f 39 ...100101 Firefox/9
00b0 31 2e 30 0d 0a 61 63 63 65 70 7a 3a 20 74 65 78 ...1.0-Acc ept: tex
00c0 74 2f 68 74 6d 6c 2c 61 70 70 69 63 61 70 69 ...t/html; applicati
00d0 6f 6e 2f 78 68 74 6d 6c 2b 78 6d 6c 2c 61 70 70 ...on/xhtml+xml,app

Part 2: Analyze a Router for Forensic Evidence

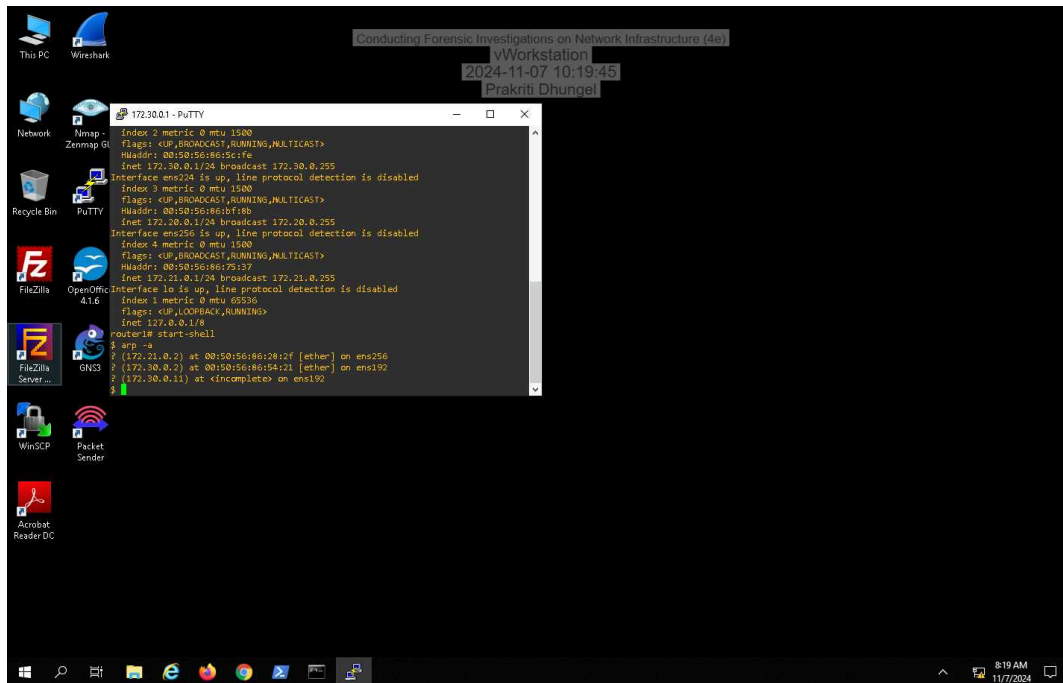
5. Make a screen capture showing the router's version output.



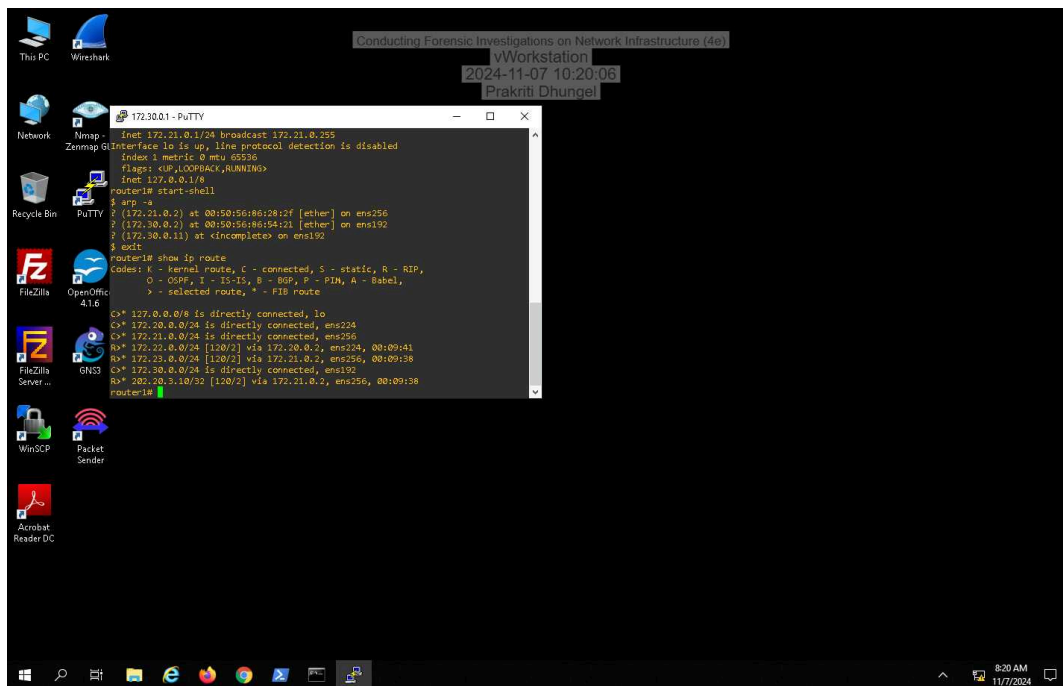
7. Make a screen capture showing the router's interface details.



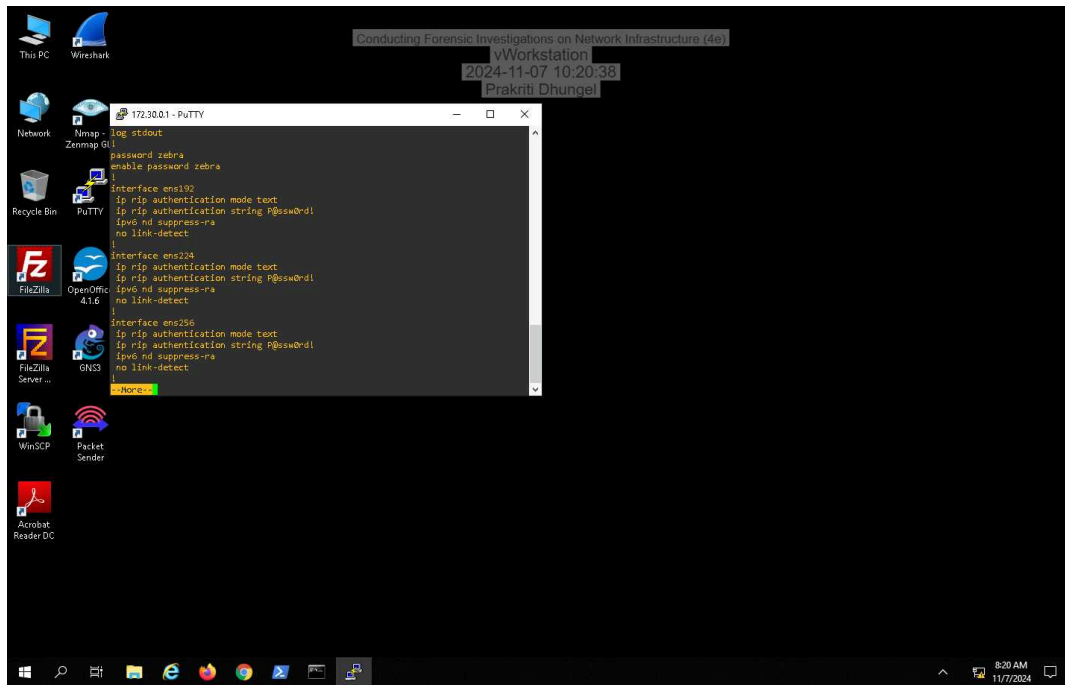
10. Make a screen capture showing the router1 ARP table.



13. Make a screen capture showing the IP routing table.



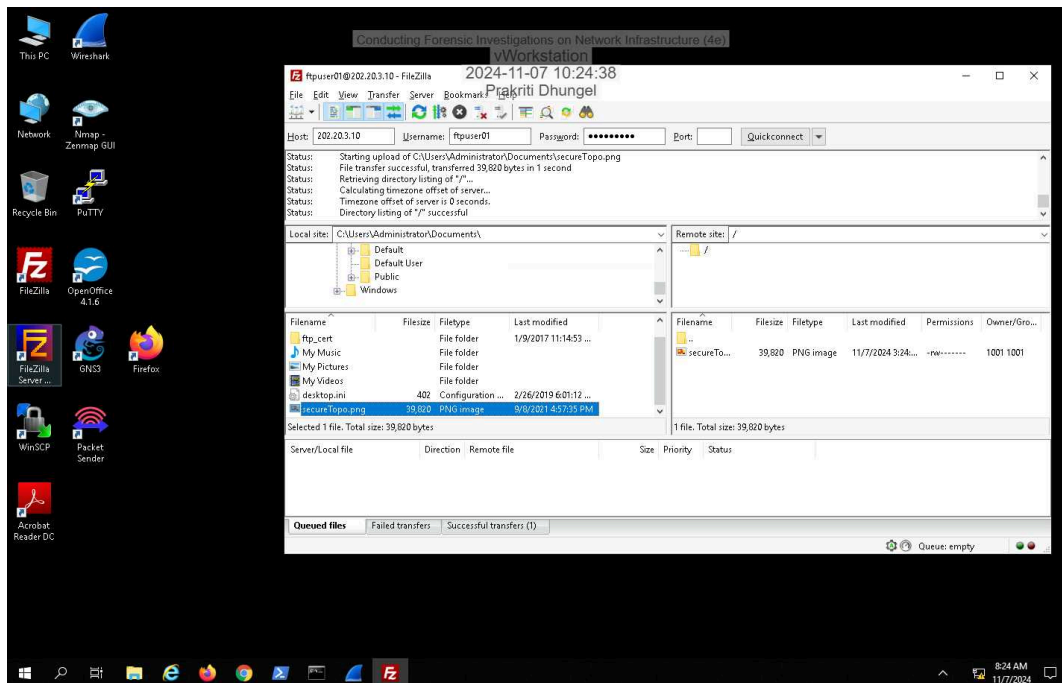
15. Make a screen capture showing the currently running configuration.



Section 2: Applied Learning

Part 1: Perform Advanced Packet Capture and Analysis

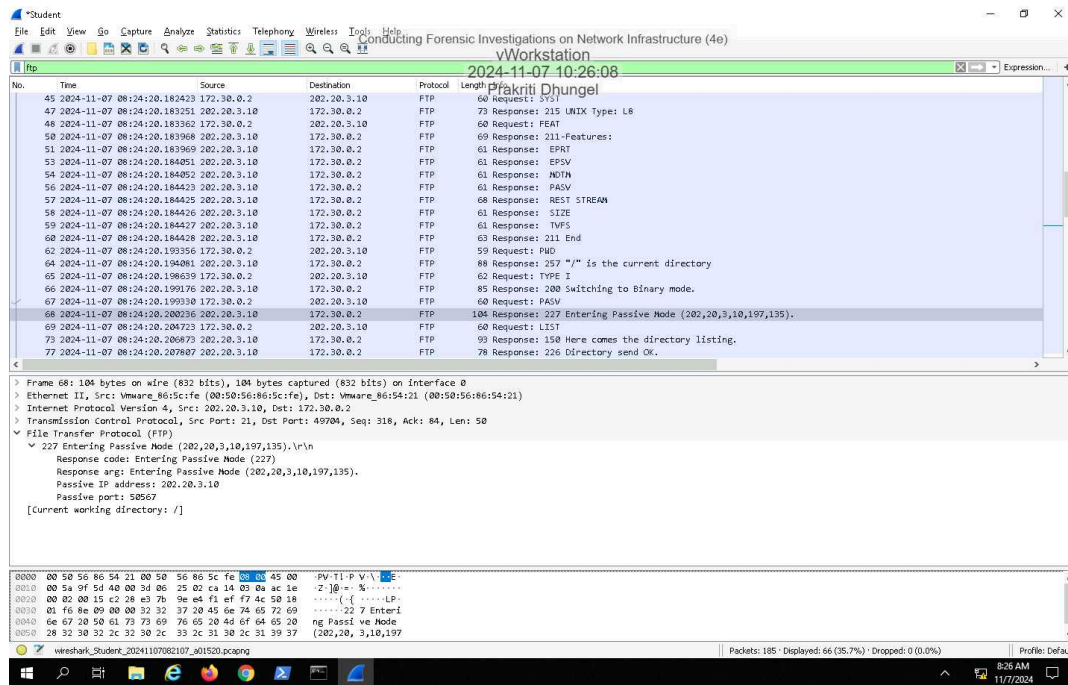
7. Make a screen capture showing the **successful transfer of the secureTopo.png file**.



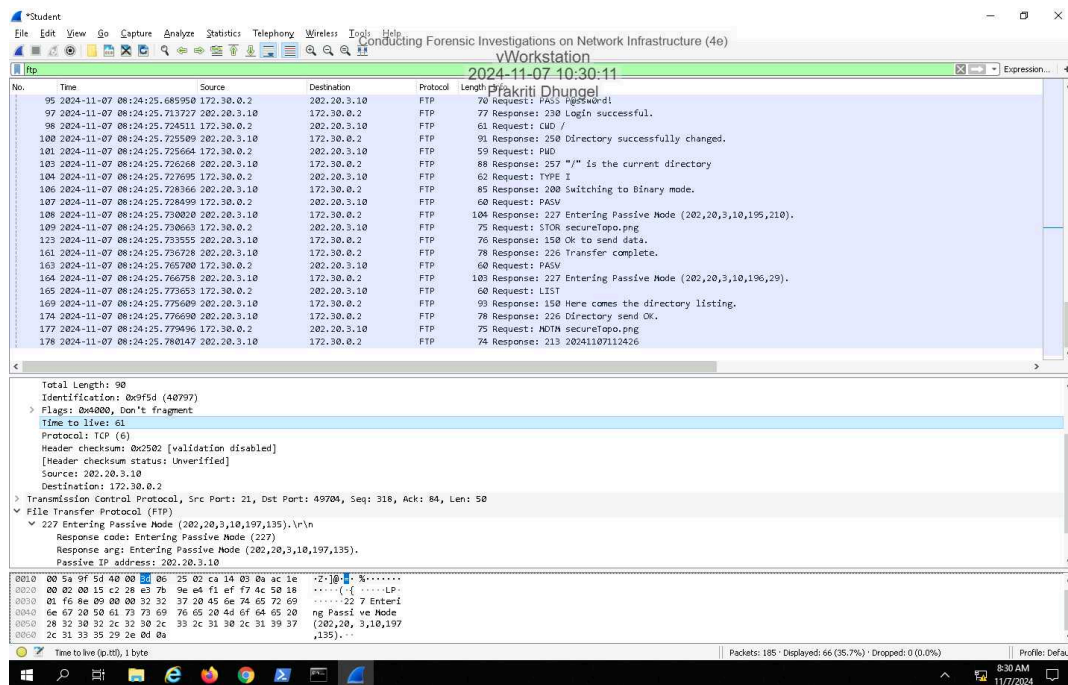
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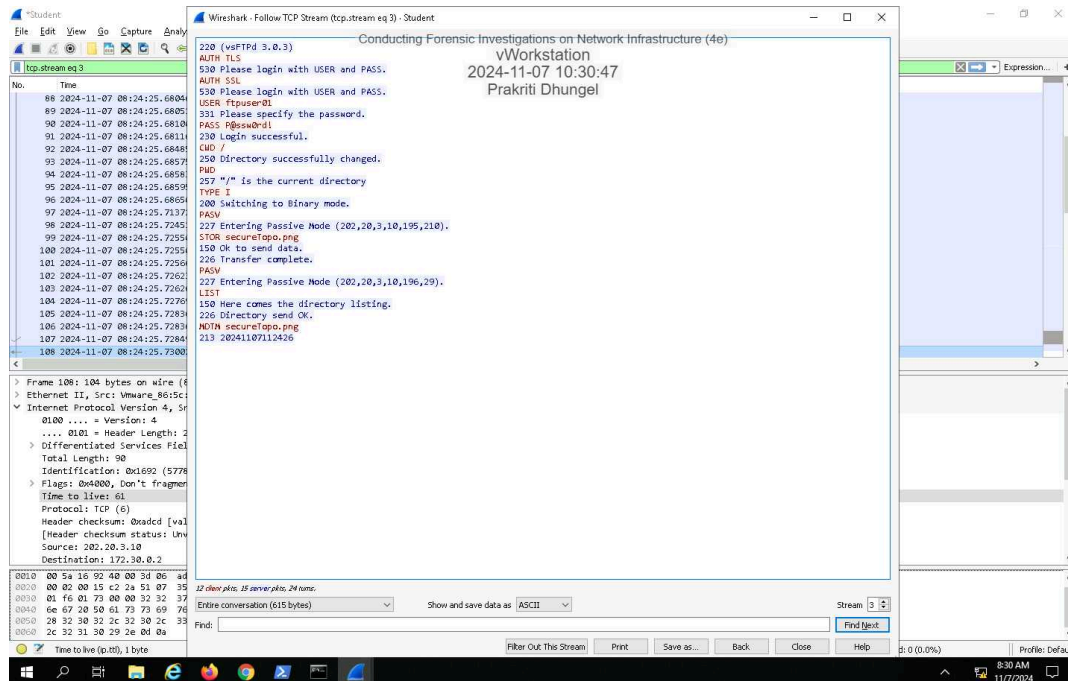
15. Make a screen capture showing the passive port specified by the FTP server in the Packet Details pane.



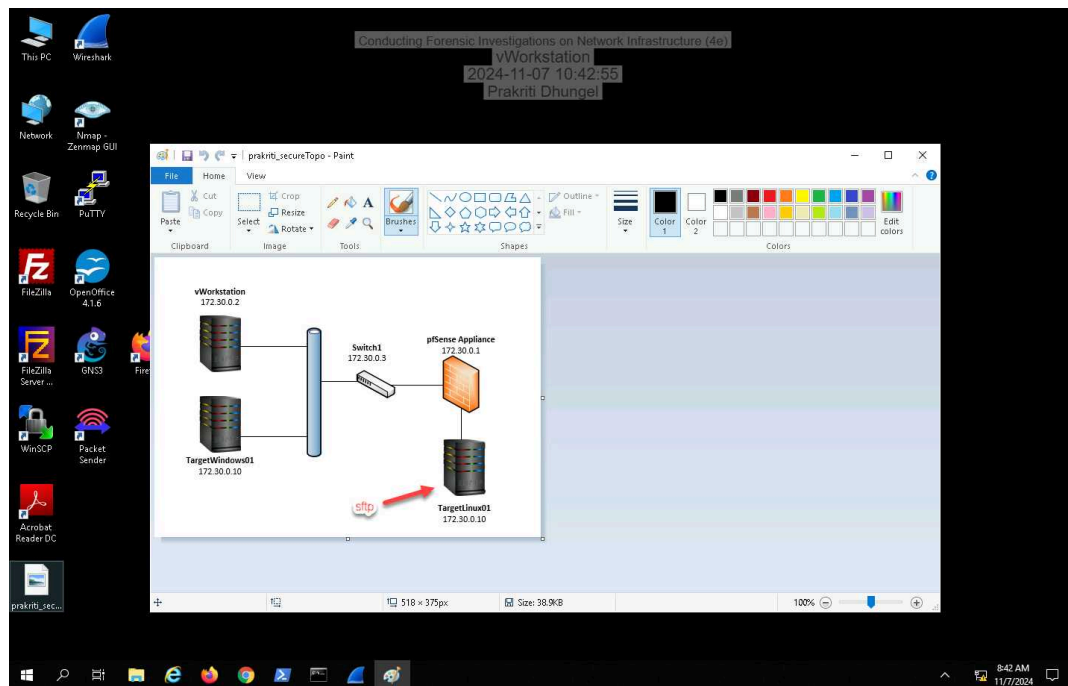
18. Make a screen capture showing the Time to live field in the Packet Details pane.



20. Make a screen capture showing the Follow TCP stream window.

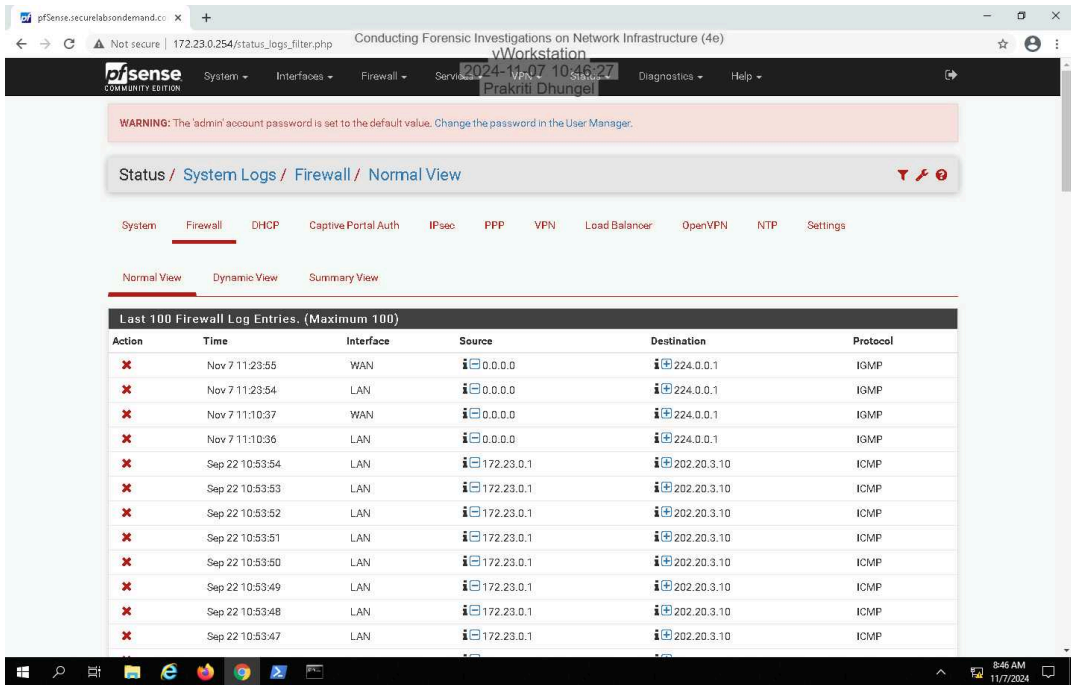


32. Make a screen capture showing the reconstituted PNG file.



Part 2: Analyze a Firewall for Forensic Evidence

9. Make a screen capture showing the entries in the firewall log.



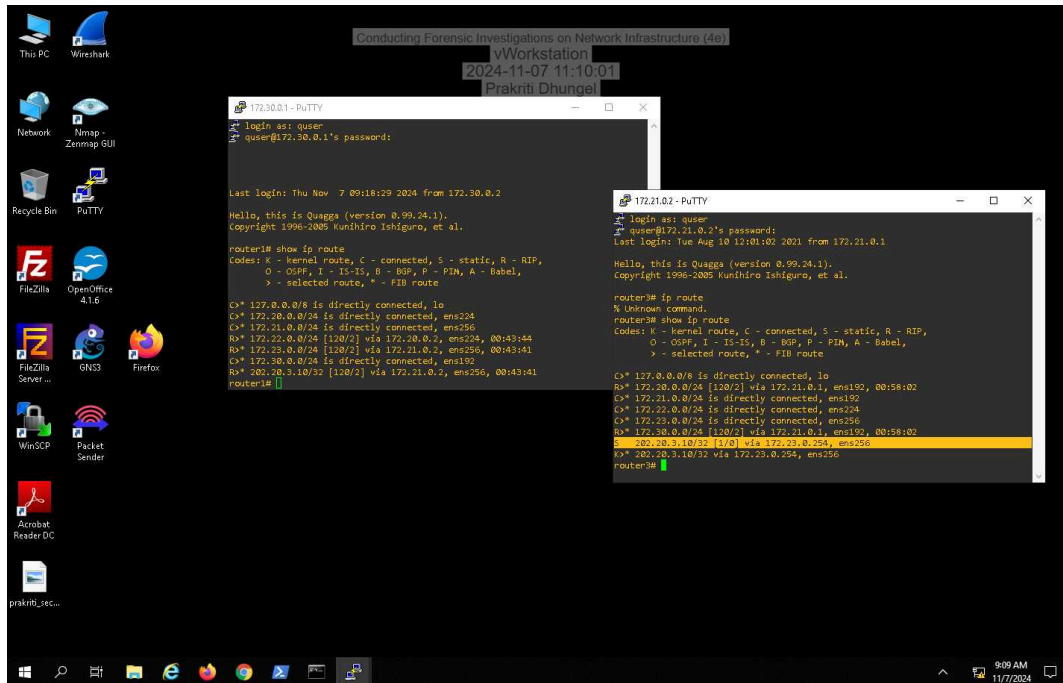
11. Make a screen capture showing the resolved entries in the firewall log.



Section 3: Challenge and Analysis

Part 1: Identify the Source of a Suspicious Route

Make a screen capture showing the non-RIP route that you discovered on the target router.



Part 2: Identify Suspicious Outgoing Connections

Record the destination IP address and Port number of the outgoing connection attempt.

Destination IP address: 202.20.3.10 Destination port number: 1337